Smart Campus Navigation Platform

Fall 2025 Deliverables Overview

Team 9

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Our Goal:

Create an integrated smart-campus mobile platform that enhances navigation, convenience, and overall student experience.

Campus Navigation Challenges

Navigating NU's campus is difficult due to fragmented information:

- New students struggle to find buildings, rooms, and services
- No unified map covering academic buildings, dorms, and campus facilities
- No real-time indoor/outdoor route guidance
- Events, services, and discounts are scattered across multiple sources
- Students lack a single, cohesive campus companion

Backend system

A comprehensive backend system for Smart Campus Navigation that provides:

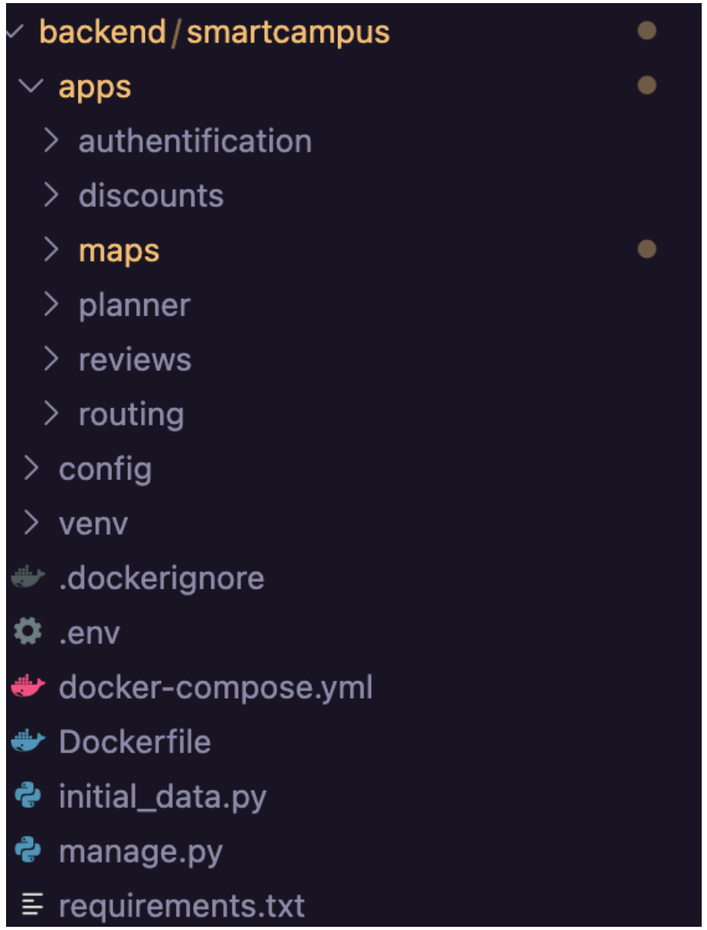
- Real-time positioning (GPS)
- Intelligent routing (Shortest, Accessible, Least Crowded)
- Student services (GPA calculator, Planner, Discounts)
- Campus information (Buildings, Rooms, Events)

Fall 2025 overview:

- 6 Django apps
- 14 database models
- 40+ API endpoints
- 25+ serializers
- 6 service classes
- 15+ test cases

Spring 2026 plans:

- Build full routing engine.
- Integrate hybrid positioning data.
- Connect university services and authentication.
- Add ML-based features into backend (sentiment + crowd prediction).
- Optimize, test, and deploy backend.



Backend system

RESTful API (40+ endpoints)

Authentication (/api/v1/auth/)

- Register, Login, Logout
- Token refresh
- Profile management

Maps (/api/v1/maps/)

- Buildings CRUD
- Rooms by building
- Nearby search
- Events & Obstructions

Routing (/api/v1/routing/)

- Calculate routes
- Reroute functionality
- Saved routes management

Reviews, Planner, Discounts (/api/v1/...)

- Full CRUD operations
- Filtering and search
- Pagination support

auth

POST	/auth/login/	auth_login_create
GET	/auth/profile/	auth_profile_list
POST	/auth/register/	auth_register_create
POST	/auth/token/refresh/	auth_token_refresh_create

discounts

GET	/discounts/	discounts_list
GET	/discounts/vendors/	discounts_vendors_list
GET	/discounts/vendors/{id}/	discounts_vendors_read
GET	/discounts/{id}/	discounts_read
POST	/discounts/{id}/claim/	discounts_claim

maps

GET	/maps/buildings/	maps_buildings_list
GET	/maps/buildings/{id}/	maps_buildings_read
GET	/maps/pois/	maps_pois_list
GET	/maps/pois/{id}/	maps_pois_read
GET	/maps/rooms/	maps_rooms_list
GET	/maps/rooms/{id}/	maps_rooms_read

planner

GET	/planner/courses/	planner_courses_list
GET	/planner/courses/{id}/	planner_courses_read
GET	/planner/gpa/	planner_gpa_list
GET	/planner/grades/	planner_grades_list
POST	/planner/grades/	planner_grades_create
GET	/planner/grades/{id}/	planner_grades_read
PUT	/planner/grades/{id}/	planner_grades_update
PATCH	/planner/grades/{id}/	planner_grades_partial_update
DELETE	/planner/grades/{id}/	planner_grades_delete
GET	/planner/progress/	planner_progress_list
GET	/planner/schedules/	planner_schedules_list
POST	/planner/schedules/	planner_schedules_create
GET	/planner/schedules/{id}/	planner_schedules_read
		planner_schedules_update
		planner_schedules_partial_update
		planner_schedules_delete

reviews

GET	/reviews/	reviews_list
POST	/reviews/	reviews_create
GET	/reviews/{id}/	reviews_read
PUT	/reviews/{id}/	reviews_update
PATCH	/reviews/{id}/	reviews_partial_update
DELETE	/reviews/{id}/	reviews_delete

routing

POST	/routing/calculate/	routing_calculate_create

AI/ML Components of our system

Fall 2025 overview

Research & Model Development

- Route optimization algorithms (multi-criteria pathfinding)
- Crowd prediction models (LSTM, temporal patterns)
- Sentiment analysis pipeline (NLP, transformer-based)
- Target: >85% precision/recall for sentiment classification

Dataset Preparation

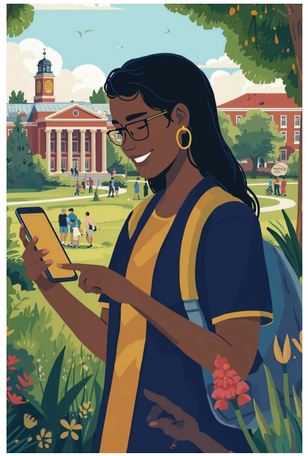
- NU-specific data collection
- Review corpus collection and labelling
- Campus graph structure for routing
- Feature engineering for ML models

Technical Foundation

- TensorFlow Lite integration research
- Data pipeline architecture design
- Model evaluation framework

Spring 2026 plans

- Train and validate all ML models
- Backend integration (APIs for predictions)
- Build analytics dashboard
- Real-time crowd prediction deployment



Organization	Discount	Details	Instagram	Address Link	Instagram Link
Inhype perfume	10%	Show your ID	@inhype_perfume	https://maps.go	https://instagram.com/
CAS beauty spa	10%	Show your ID	@cas_beauty_spa	https://maps.go	https://instagram.com/
Safar Saqtau	15%	Show your ID	@safarsaqtau	https://maps.go	https://instagram.com/
Amari Garden	15%	Show your ID	@amari.garden.flowers	https://maps.go	https://instagram.com/
Veggie House	10%	Show your ID	@veggie_house01	https://maps.go	https://instagram.com/
Arti laser	25%	Show your ID	@arti_laser	https://maps.go	https://instagram.com/
BRO GLASSES	15%	Show your ID	@broglasses.astana	https://maps.go	https://instagram.com/
Jasyl coffee	10%	Except delivery, only for	@jasylcoffee	https://maps.go	https://instagram.com/
Teadot	10%	Show your ID	@teadot_astana	https://maps.go	https://instagram.com/
Moon Collection	10%	Until the end of may. Sho	@moon.collection.kz	https://maps.go	https://instagram.com/
Focus Telo	10%	Show your ID	@focustelo	https://maps.go	https://instagram.com/
Lammi Me	10%	Free brow lamination on	@lammi.me	https://maps.go	https://instagram.com/
Dodo Pizza	10%	10% discount on online	@dodopizza_astana		https://instagram.com/
Essence	10%	Does not apply on indivi	@essence.ast	https://maps.go	https://instagram.com/
O'sima	10%	Only on Thursdays. Show	@osima_cosemics	https://maps.go	https://instagram.com/
Sienna flowers	10%	Show your ID	@siennaflowers.kz	https://maps.go	https://instagram.com/
GOAT Cyber Club	15%	Show your ID	@goat.cyberclub	https://maps.go	https://instagram.com/
Saladbar	10%	Always, except lunch co	@saladbar.kz	https://maps.go	https://instagram.com/
Party Centre TSE	10-15%	Discount depends on th	@party_centre.tse	https://maps.go	https://instagram.com/
Honey Flowers TSE	10-15%	Discount depends on th	@honey_flowers.tse	https://maps.go	https://instagram.com/
Elite Sushi	10%	From Monday to Thursda	@elite_sushi_astana		https://instagram.com/
Bento Astana	10% big ca	Show your ID and use pr	@bento.astana		https://instagram.com/
SAVOA club	15%	Show your ID	@savoaclub		https://instagram.com/

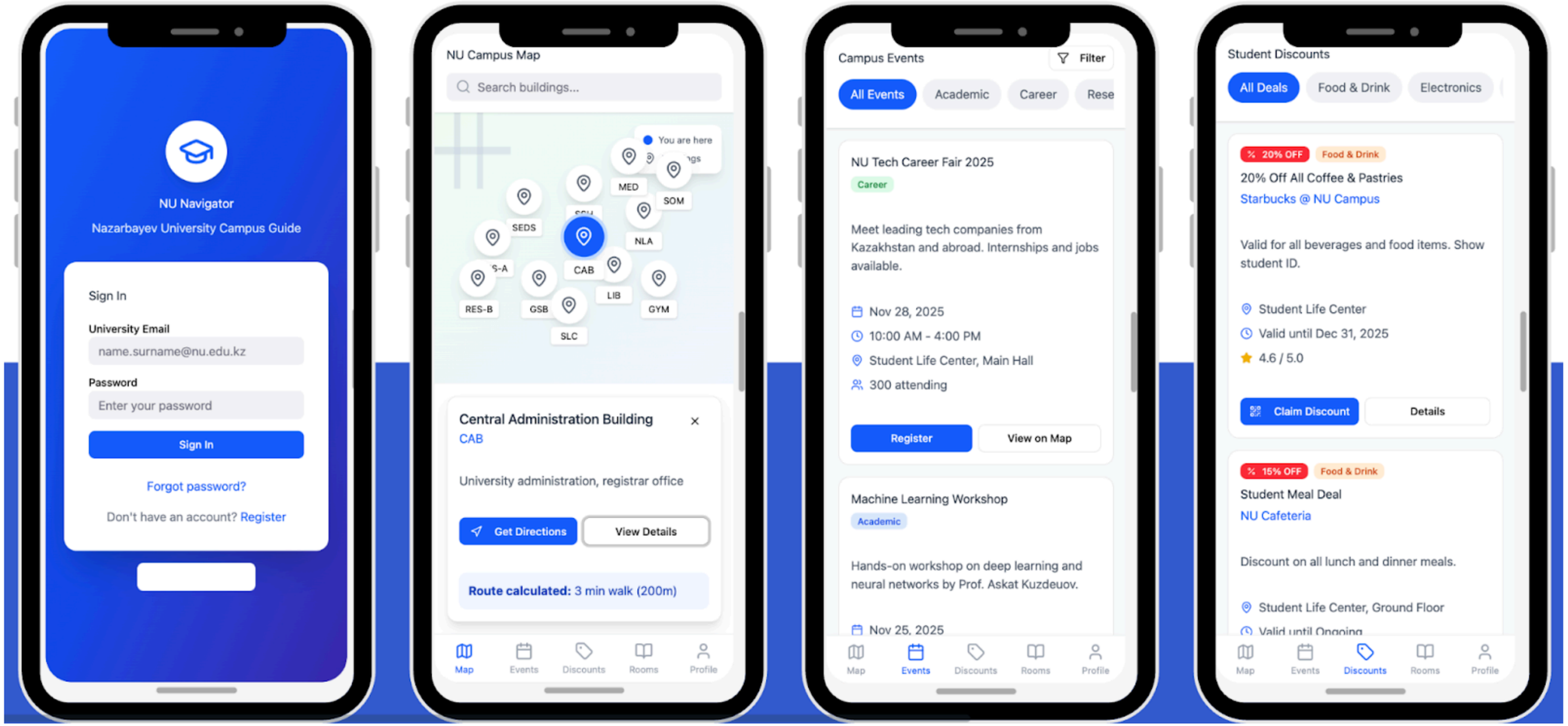
Discounts dataset

Challenges

- Limited real campus usage data
- Indoor map constraints and accuracy
- Balancing model complexity with mobile performance

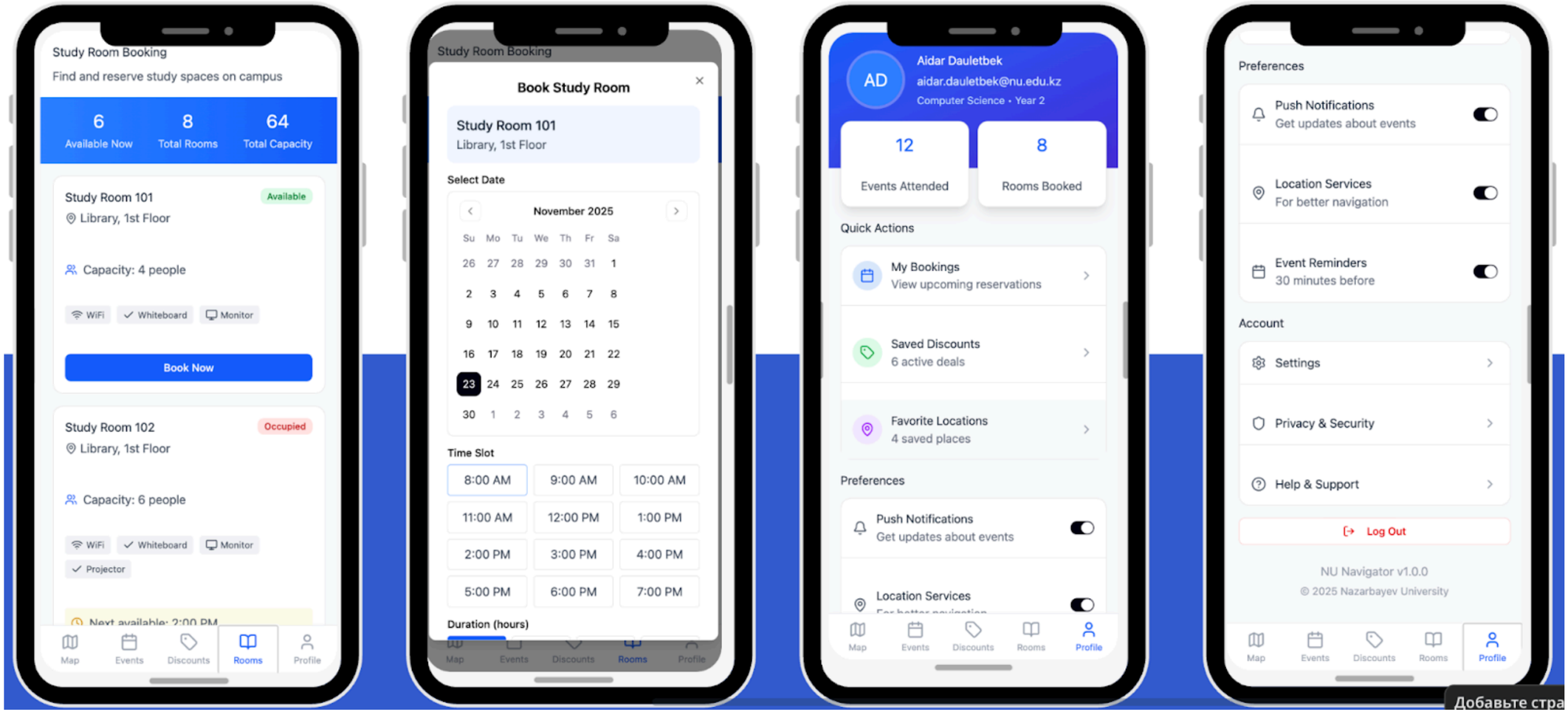
Data Pipeline Architecture

Data Collection (GPS, reviews, schedules) →
Preprocessing & Feature Engineering →
ML Model Training/Inference (TensorFlow) →
Results Storage (PostgreSQL) →
REST APIs for real-time predictions



Design Overview

Login & onboarding screens for quick access
A campus map with building details and route options
Events feed combining academic and student-life activities
A Student Deals section with curated discounts and offers



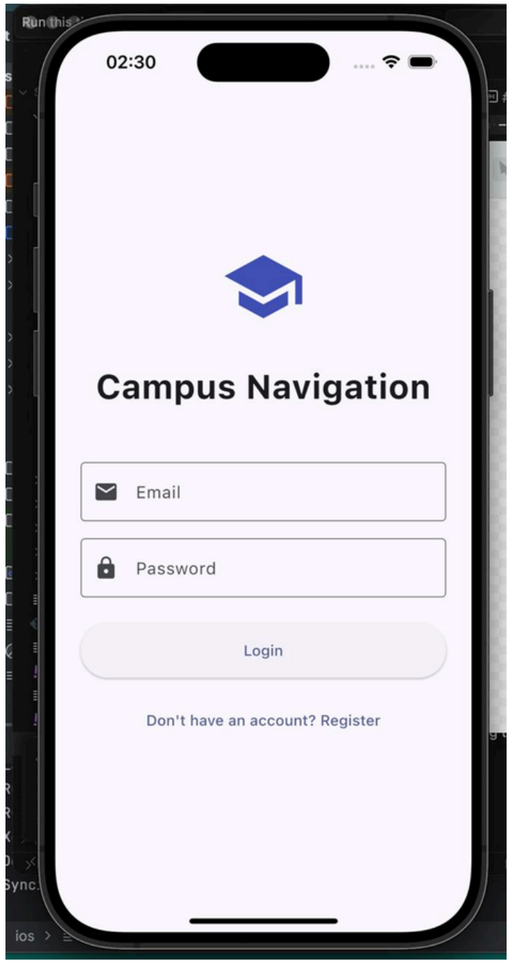
Design Overview

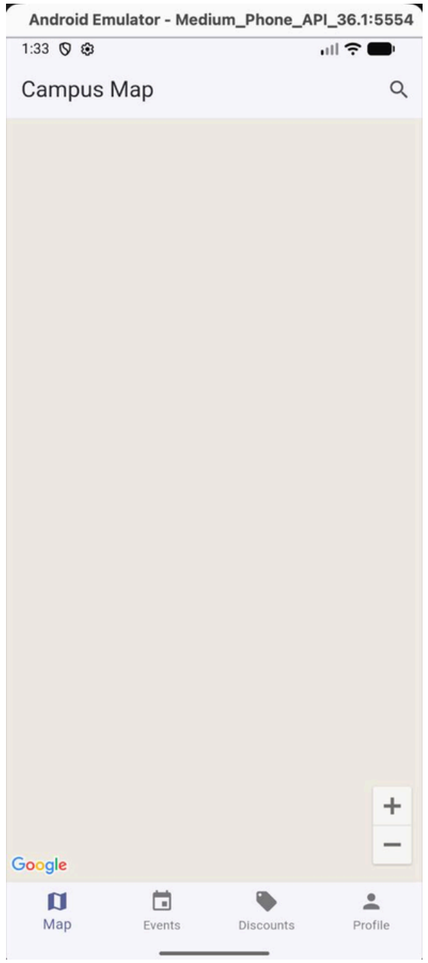
Room Booking

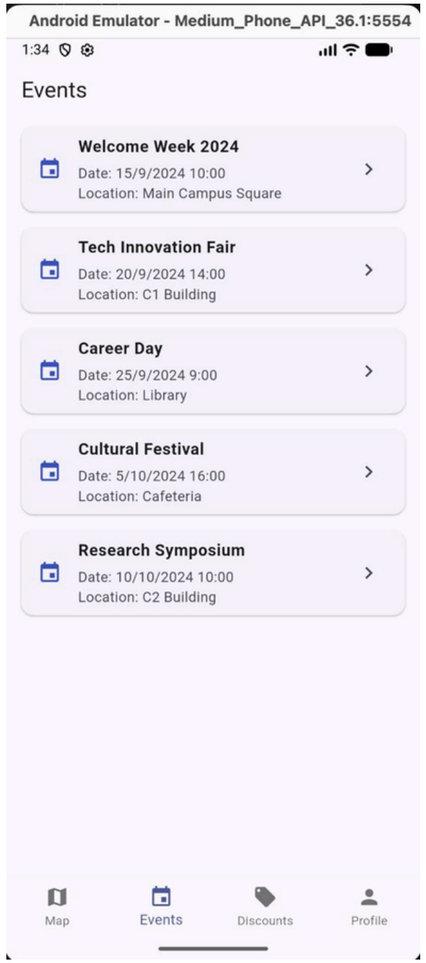
Calendar View: A built-in scheduling page

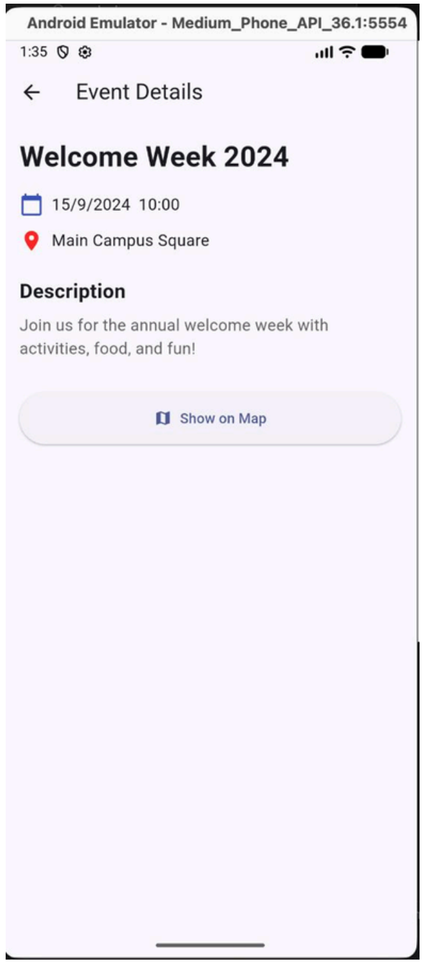
User Dashboard: A simple overview of upcoming events

Preferences & Settings: Controls for notifications, location services.



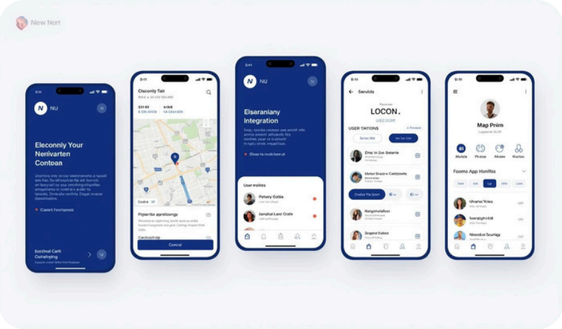






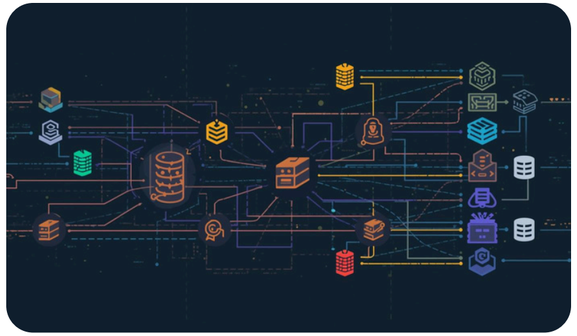
The demo version of the mobile app

Expected Deliverables for Spring 2026



Full App

Complete integration of all features and functions.



Backend

Robust infrastructure to support user demands.

Artificial Intelligence

Enhanced features for personalized user experiences.

Augmented Reality

Interactive navigation and immersive information delivery.